

Amazon E-commerce

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Abstract— Amazon.com, Inc has become one of the largest companies in the world within the last decade. Although it started as an online bookstore, it quickly transitioned into a successful e-retailer within billions in net sales. Amazon’s most popular aspect is its e-commerce which is used by millions of people daily. Its practice in ensuring convenience for its customers ensures that most customers are loyal to Amazon’s services.

1 PROBLEM SPACE

1.1 Problem and Context

Currently, there are approximately over 230 million people being Amazon customers within the United States. Amazon is such a popular service that there are billions in sales per day (Mosby, 2024). Nevertheless, despite Amazon’s overwhelming popularity there are many clear problems throughout the services that it offers. One major issue that I have recently run into when using this site is the inability for delivery that came up based on location. Whenever a user orders a product off of Amazon’s site, the customer must enter in his or her address for delivery. However, since Amazon utilizes its own personal location services to locate the address, there can sometimes be discrepancies within the results of the address.

Address Verification

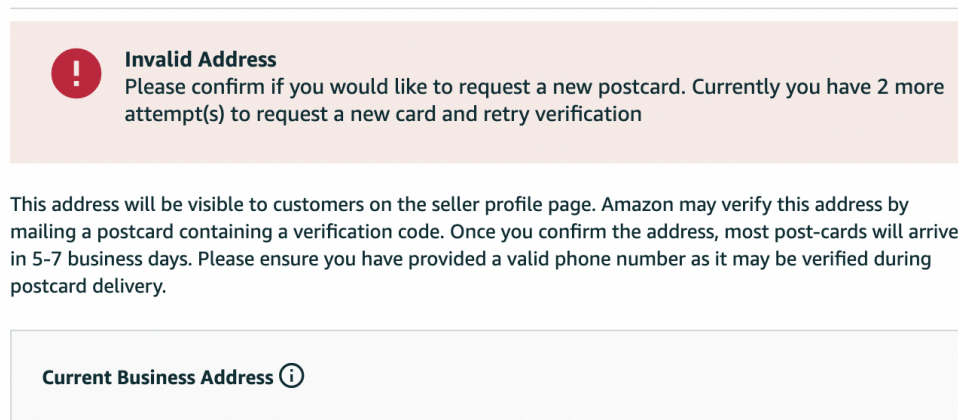


Figure 1— Image of Amazon’s Invalid Address. [Source](#)

The image above is the standard response from Amazon when entering in an invalid address that Amazon was able to identify properly. However, the situation that I personally experienced is where the delivery address was able to be verified, but the delivery itself was not able to be completed. Since the building that I had my package delivered to was a newly built building with a newly generated address, Amazon was able to have the address confirmed, but when it came to delivery, the package was not able to be delivered properly. The issue stems directly from Amazon's personal Amazon Location Service not being as completely updated to date as other location services. Although this issue is fairly miniscule for Amazon's large e-commerce network; however, if this issue is consistent throughout all newly built buildings then Amazon's profit will take a hit from the profit it can be making from customers in newly built buildings, which are customers who tend to buy a lot of products when first moving in.

1.2 Proposed Design

The entire process of Amazon's order and delivery service is extremely fast paced and smooth. My proposed design to continue this type of streamlined processing of goods revolves around two separate points to fully correct the issue involving new addresses in the navigation of deliveries. The first section involves direct manipulation of Amazon's Amazon Location Service. The Amazon Location Service algorithm to locate an address involves using geocoding. The user gives the service an address in some form or another and will receive latitude and longitude in exchange. This will then allow the user to locate the address (Places Search - Amazon Location Service, 2024).



Figure 2— Image of Amazon Location Service Geocoding. [Source](#)

Although Amazon Location Service has a built in way to locate new addresses, this process does not cover enough cases for it to be viable. My design adds in an additional check with Google's open source Maps Embed API to double check whenever an address given by the user is marked as invalid. Since Google's Maps are consistently updated every few days, Google's Maps are most likely to be more accurate than Amazon's Location Service database.

The second addition I would make to the design to resolve the invalid address issue would be the addition of a location address update request page to Amazon's website so that user's will be

able to put in requests to add in newly created addresses so that Amazon Location Service can become more accurate. Currently, the only way that users are able to request an address check or update would be to call directly into customer service to get this issue resolved, but can not always be reliable. The user needs to have a way to directly add in new addresses that Amazon may not know about yet.

1.3 Comparison

Both of the additions in the redesign of Amazon will allow Amazon deliveries to become more accurate and proficient than it already is. These two changes should be able to handle this edge case in its deliveries that Amazon is encountering. Ultimately, Amazon's delivery will be perfected even further so that it can gain even more loyal customers.

2 NEED FINDING EXERCISE

2.1 Need Finding

Amazon Location Services currently takes in an address and uses the addresses' multiple attributes to find a match in the place index. These attributes include street number, street, city, region, postal code, etc. The more attributes an address contains the more accurate a match in the index can be found. New addresses are currently handled by interpolation of previously known addresses to determine location.

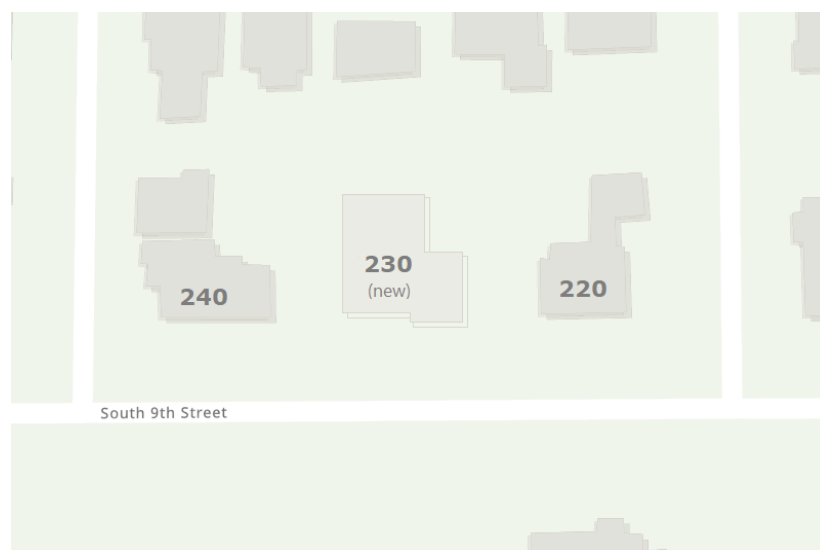


Figure 3— Image of Amazon Location Service Interpolation. [Source](#)

In this example of Amazon's interpolation, the 230 address of South 9th Street is a newly created building. The location service is able to determine the location of the building based on the previously used address of 220 South 9th Street and 240 South 9th Street, even though 230 is not

within the known database. However, if there is little to no known location near the new address then this method will not work and the address will be identified as invalid.

2.2 Functional Components

This redesign will include the use of Google's Maps Embed API as an additional check to Amazon Location Services' address result. The Google API will be utilized only on the condition that ALS returns a result of invalid so that this result can be cross checked with Google Maps so that newer buildings aren't being excluded from deliveries due to newer addresses. Google Maps' is one of the most updated maps available that is constantly being updated every few days. This ensures that the Maps API can give the most accurate response on invalid address checks.

2.3 Architecture

Pseudo Code/FlowChart for Amazon Location Services with Google Maps API

Amazon Location Service API

Enter street address (at least one)

Street number, street, city, region, postal code

ALS API Geocoding

Interpolation returns false or true

If true

Returns known address location on map

If false

Google Maps Embed API with

src ="https://www.google.com/maps/embed/v1/place?key=API_KEY"

And q = street address

If no address found

Address is invalid

2.4 Data Collection

For this design, the data collection is heavily dependent on the results of correctly returned addresses. This requires the utilization of newly built buildings to test whether or not the addition

of a Google Maps check on the deliveries are able to screen out the false positives from the truly invalid addresses. The results of how many false positives were able to be caught will determine how effective this redesign will be. Since this solution is completely dependent on Google Maps regular updates, it is not possible to completely get rid of false positives in the invalid addresses; however, this should assist in minimizing a large portion of these deliveries.

2.5 Utility Types

The main goal of this design change is the improvement in functionality of the navigation location service that Amazon utilizes in its delivery services. If a delivery service is not able to deliver its packages to its customers properly then there should be no trust in the company at all. This design aims to correct any invalid address that Amazon Location Service tends to have so that everyone will be able to get proper deliveries on products.

3 DESIGN REFINEMENTS

3.1 Need Finding

The previous design alterations focused on correcting the issues that Amazon Location Service had within its navigation service. However, this redesign will revolve around including a way for users to help reduce invalid addresses.

3.2 Functional Components

Google and many other navigation location services have already implemented ways that allow users to be able to request addresses to be added to their maps or change addresses as a whole.

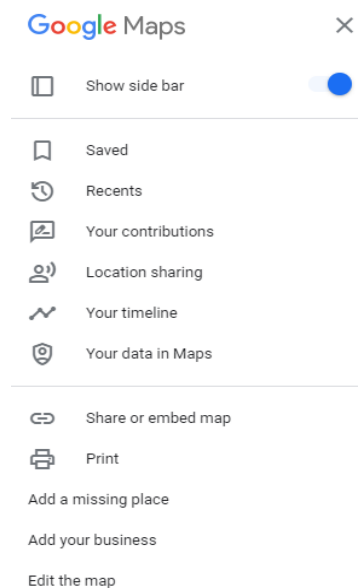


Figure 4— Image of Google Maps' taskbar. [Source](#)

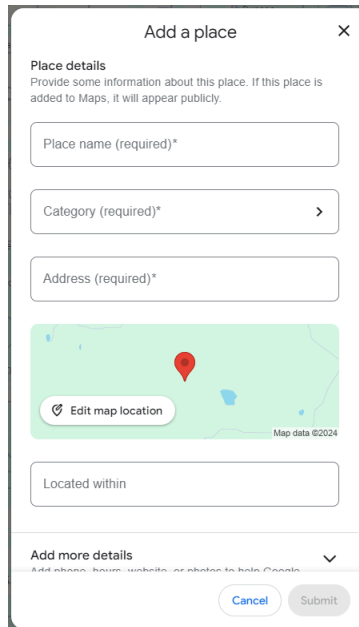


Figure 5— Image of Google Maps’ Add a Missing Place Screen. [Source](#)

This redesign will include a section similar to Google’s map functionality to add a missing place. This design will include a place’s street number, street, city, and postal code as the minimum so that this address can be confirmed by Amazon. This will give users another way of adding in missing addresses in Amazon without having to call into the customer service line.

3.3 Architecture

This address verification request page will be placed within Amazon’s shipping and delivery help page so that users will be able to see this tool whenever issues with their deliveries arise.

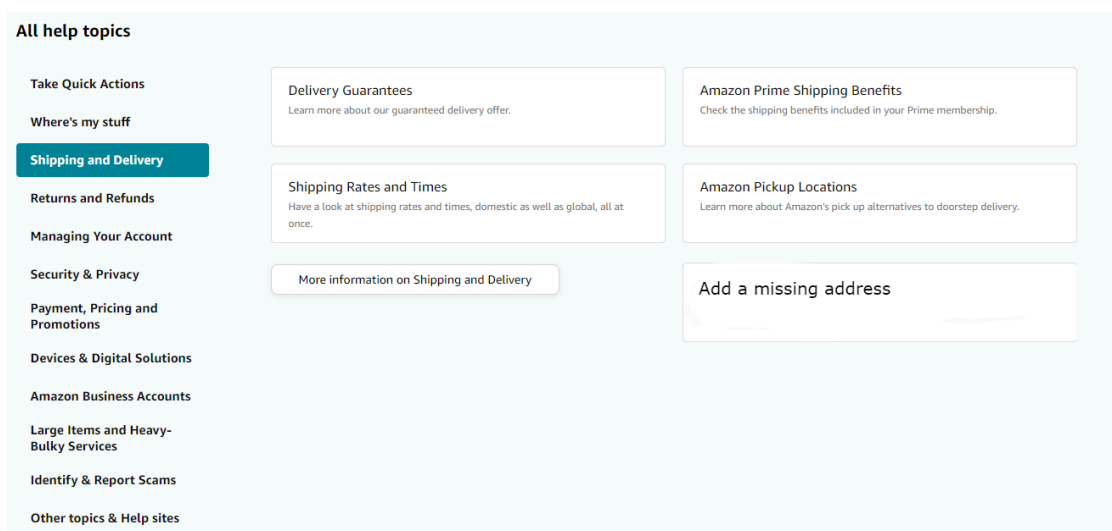


Figure 6— Image of Amazon Help Screen with new Button Design

3.4 Data Collection

This redesign revolves around receiving requests from users to assist in the management of Amazon's place index resource. As long as Amazon is able to take the addresses the user requests and confirm whether or not the new addresses are real or not, these addresses can be added to the database so that this new delivery address can be functional. The data collected from this design will be solely analyzed from the number of user requests and the number of real addresses added to the database. Both of these can help determine whether or not this addition to Amazon's current site could help deliveries or not.

3.5 Utility Types

This task has the intended use of resolving any possible false positive invalid addresses that may come up for users. As long as users are able to input their addresses and have it confirmed by Amazon as soon as possible, their addresses will be able to be added to Amazon's database. This will help reduce any possible invalid addresses from new buildings since these addresses will have already been recorded.

4 ADDITIONAL DESIGN CONSIDERATIONS

4.1 Evaluation of Baseline

For the baseline approach, the evaluation and analysis of the designs are determined through the number of successes found in the data collected. The data recorded in the designs all identify whether or not an address is ultimately added into the place search index that Amazon uses for its location services. The number of positive additions to this index will determine whether or not these designs were successful. In addition, low numbers of additions to the database does not necessarily mean that the designs were unsuccessful. The context revolves around new buildings being built with new addresses that are not known. Each development takes time and so the success or failure of these designs must be analyzed over a certain length of time to determine if it is truly successful or not.

4.2 Additional Requirements

Additional requirements for the designs will be heavily geared towards functionality for the first addition and simplicity for the second change. The first change involves the actual functionality of the navigation location service within Amazon's delivery service so the accuracy of each component is crucial. The navigation for package delivery is a major part of Amazon's e-commerce so the delivery functionality needs to be ensured. Especially for the edge case of new addresses causing invalid addresses for deliveries. The second design change involves user input to assist with resolving the overall invalid delivery issue before it gets to the delivery stage.

This design involves the creation of a new page so that users can add a new address to Amazon's place search index, so this component requirement will be simple. The users need to have a simple and easy way to request a new address be confirmed and added so that deliveries can be done properly. This encompasses all the additional requirements that each component of the design needs to have to assist the overall system performance.

5 REFERENCES

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2. *Places search - Amazon Location Service*. (2024). Amazon.com. <https://docs.aws.amazon.com/location/latest/developerguide/places-concepts.html#::text=Addresses%20%E2%80%93%20a%20search%20for%201600>
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